MILITARY SPECIFICATION SHEET

# CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, PIN CONTACTS, 160 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS, (.100 INCH SPACING)

This specification is approved for use within Rome Air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

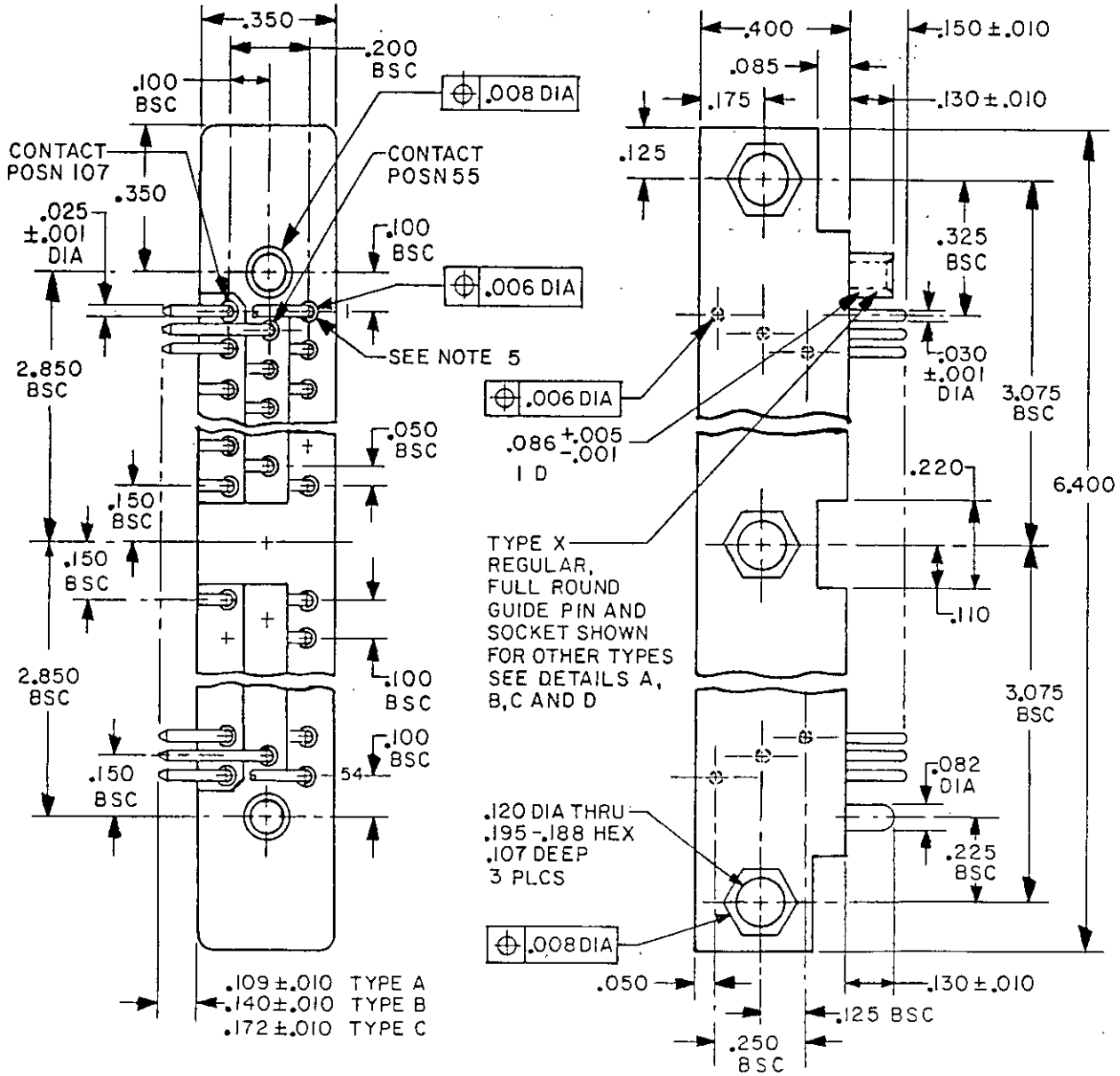
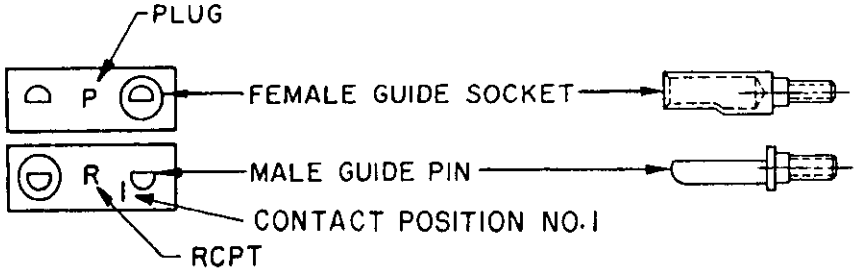


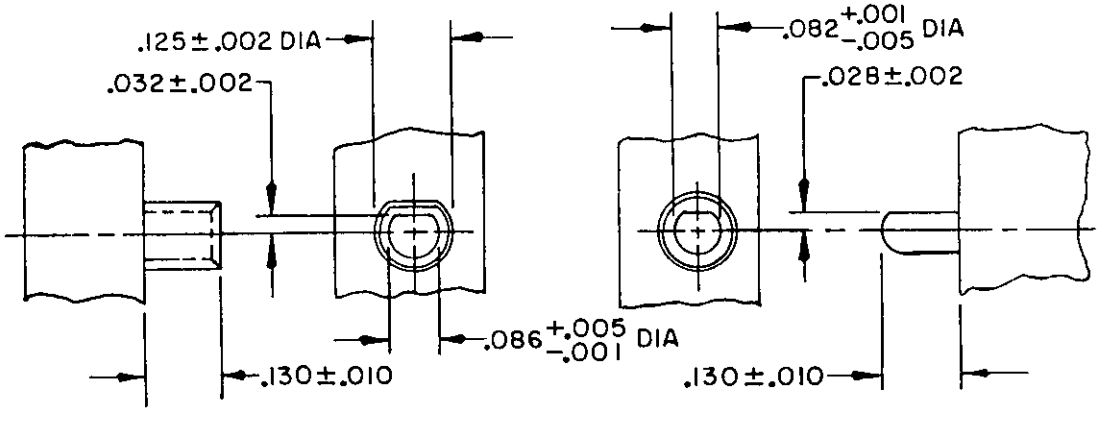

FIGURE 1. Connector, plug .100 inch (2.54 mm) spacing.

(A) denotes changes


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CONNECTOR POLARIZATION WITH D SHAPED GUIDE PINS AND SOCKETS. (TYPE Y)

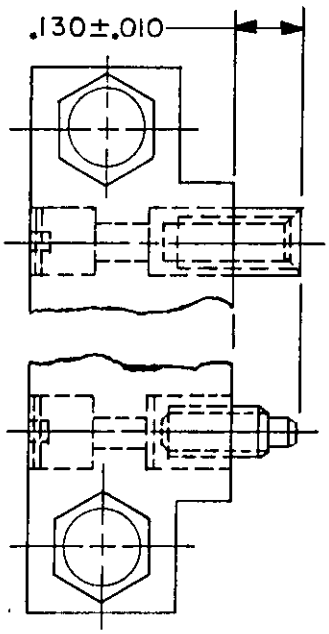



DETAIL A

FIGURE 1. Connector, plug .100 inch (2.54 mm) spacing - Continued.

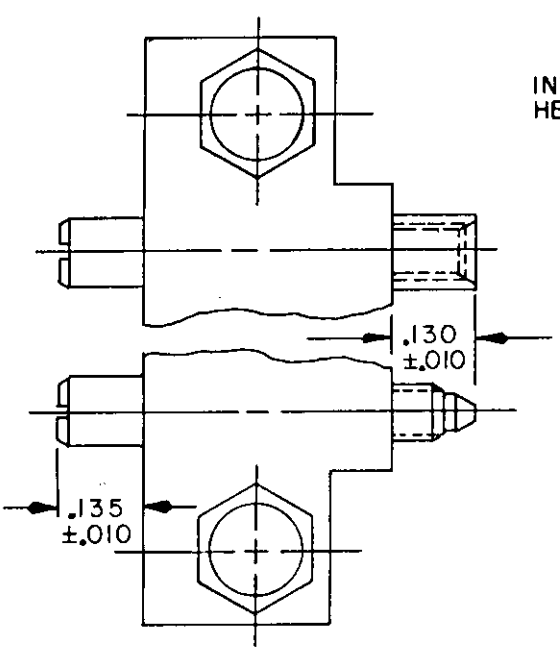


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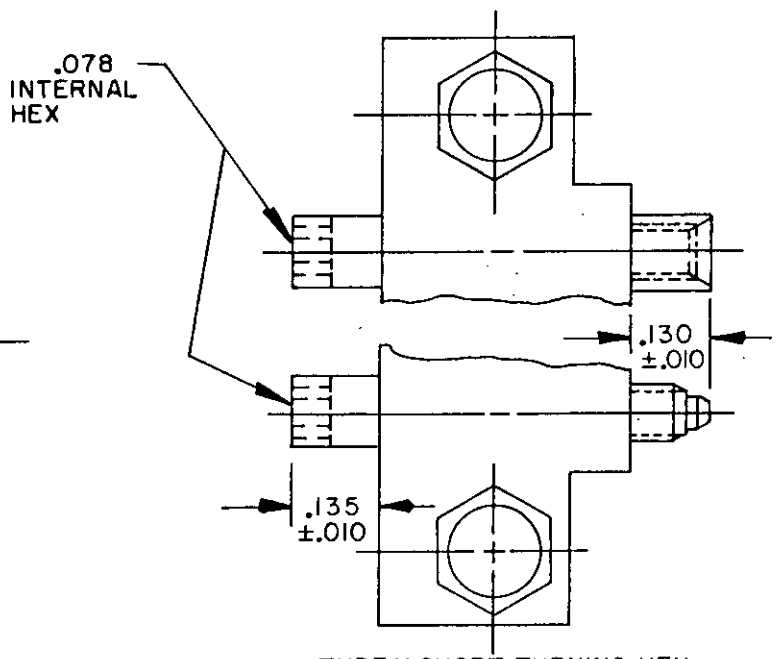

TYPE F FIXED JACKSCREWS

DETAIL B




TYPE S SHORT TURNING SLOTTED
JACK SCREW

DETAIL C




TYPE H SHORT TURNING HEX
JACK SCREWS

DETAIL D

FIGURE 1. Connector, plug .100 inch (2.54 mm) spacing - Continued.



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| Inches | mm | Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|---|---|
| .001 | 0.03 | .032 | 0.81 | .109 | 2.77 | .172 | 4.37 | .325 | 8.26 |
| .002 | 0.05 | .050 | 1.27 | .110 | 2.79 | .175 | 4.44 | .350 | 8.89 |
| .005 | 0.13 | .078 | 1.98 | .120 | 3.05 | .188 | 4.78 | .400 | 10.16 |
| .006 | 0.15 | .082 | 2.08 | .125 | 3.18 | .195 | 4.95 | 2.850 | 72.39 |
| .008 | 0.20 | .085 | 2.16 | .130 | 3.30 | .200 | 5.08 | 3.075 | 78.10 |
| .010 | 0.25 | .086 | 2.18 | .135 | 3.43 | .220 | 5.59 | 6.400 | 162.56 |
| .025 | 0.64 | .100 | 2.54 | .140 | 3.56 | .225 | 5.72 | | |
| .030 | 0.76 | .107 | 2.72 | .150 | 3.81 | .250 | 6.35 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).
4. These connectors mate with connectors specified in MIL-C-55302/139.
5. Numbers indicating the first and last contact on each row shall be stamped on the side opposite the board mounting side. Every fourth contact shall be stamped on the first row only starting at the number one (1) contact with the exception that the 53 contact position may not be marked.
6. Termination layout on .020 (.51 mm) modular grid.
7. Optional undercut .020 (.51 mm) maximum, on PCB side for cleaning purposes.

FIGURE 1. Connector, plug .100 inch (2.54 mm) spacing - Continued.



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REQUIREMENTS

Dimensions and configuration: See figure 1.

Material: Guide pins, type X guide sockets, jackscrews and jack sockets shall be made of corrosion resisting stainless steel per ASTM-A582 passivated per QQ-P-35 or shall be corrosion resistant copper nickel alloy (61 ±2 percent copper, 25 ±2 percent nickel, 2.5 percent maximum other, and the balance zinc) and shall have a brinell hardness of 145 to 175. Type Y guide sockets shall be beryllium copper per QQ-C-530, nickel plated per QQ-N-290, 30 to 150 microinches.

Ⓐ Plating: The contact plating shall be per MIL-C-55302, or gold per MIL-G-45204, type II, grade C, class 1 over nickel plating per QQ-N-290, 30 to 150 microinches in the engaging area, and gold per MIL-G-45204, type II, grade C, class 00, over nickel plating per QQ-N-290, 30 to 150 microinches in the termination area.

Ⓐ Nonfunctional areas: See MIL-C-55302, paragraph 3.3.3.4.3, section C.

Contact identification: See figure 1.

Contact rating: 5.0 amperes maximum per contact.

Contact resistance: The average contact resistance of all contacts measured shall not exceed .010 ohm, and no individual contact pair shall have a resistance exceeding .020 ohm.

Dielectric withstanding voltage:

Sea level: 1,000 volts rms.
High altitude: 300 volts rms.

Polarization: Type X, regular, guide pin and socket, full round (see figure 1). Type Y, adjustable D-shaped guide pin and/or socket (see MIL-C-55302/57, detail B). All connectors are supplied in the -1 polarized position. When guides are removed and reinstalled in another position apply loctite 83-CV or equivalent to the threads. Ordering connectors in other than the -1 polarized position may be done by the original equipment manufacturer by placing the desired polarized position in brackets three spaces after the part number on the purchase order. Example: M55302/138-A10Y [-32]. Polarization pins and sockets shall be adjusted to any desired configuration by spanner wrench M55302/57-01. Ordering connectors in the -1 polarized position without loctite 83-CV or equivalent applied may be done by placing a "-0" in brackets three spaces after the part number on the purchase order.

Note: Hardware with Locktile 83-CV or equivalent applied shall withstand a torque of 7 in-oz applied through the nut.

Marking:

Connector part number: See example.



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Example of part number:

M55302/138 A 160 X

Basic number of specification sheet

Type of terminal (see figure 1)

A = Dip terminal .109 inch length
B = Dip terminal .140 inch length
C = Dip terminal .172 inch length

Number of contacts (see figure 1)

Type mounting hardware (see figure 1, details A, B, C, and D)

X = Full round
Y = D-shaped
F = Fixed jackscrew
S = .135 slotted turning jackscrew
H = .135 hex turning jackscrew

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 85, 99

Jser activity:
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5935-F268-1)



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